TIC 395051777

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        3.633071 d
  Epoch (t0)    1598.5123 BTJD
  Duration      0.0840 d
  Depth         3.631 %  (36,307 ppm)
  SNR           232.4

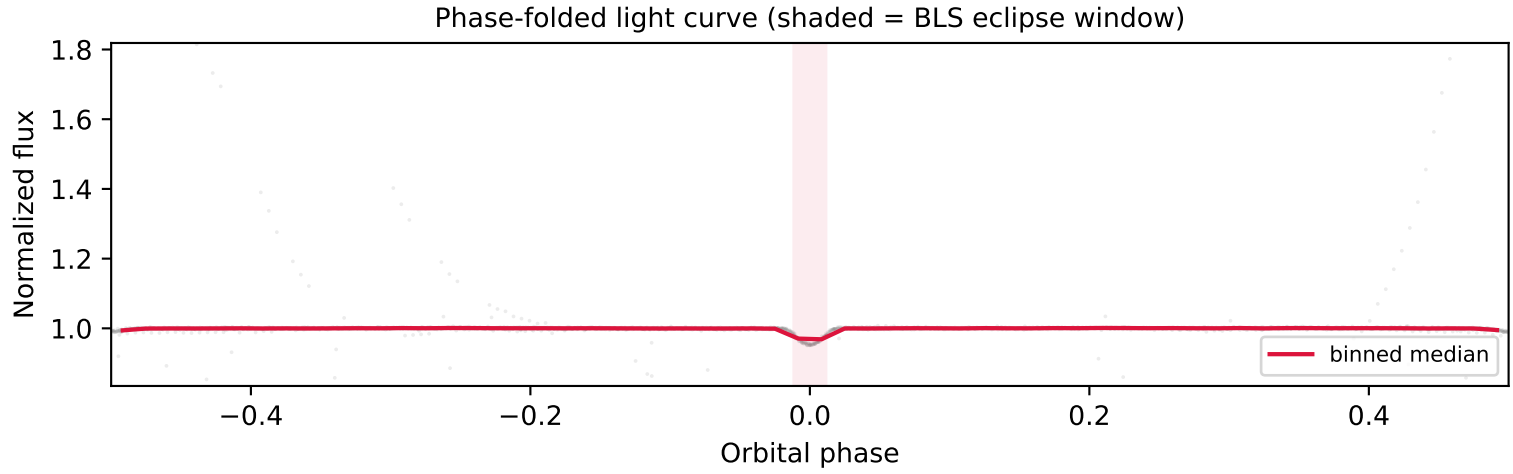
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      21  (gate >= 3)
  Robust depth ratio     1.08  (gate >= 0.50)
  Robust depth sigma     164.5  (gate >= 7.0)
  Odd/even z-score       0.11  (consistent)
  Secondary eclipse      detected, phase 0.49, 6.4 sigma
```

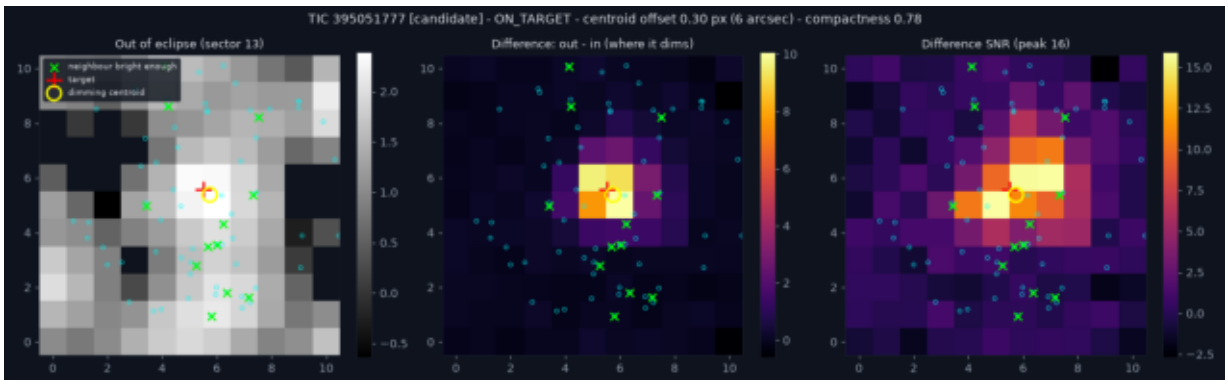
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 13    light-curve dip in this sector: 95.8 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   15.93   (gate >= 5.0)      Compactness   0.78   (gate >= 0.50)
  Dimming centroid offset     0.30 px (6 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 395051779
    1.17 px away, dTmag +2.52 - farther from the dimming than the target
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No caveats - clean detection on every check.